Endpoint Security Monitoring & Threat Intelligence Integration with Wazuh

Internee.pk — Project Report

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Report purpose
This editable report documents the recorded controlled-lab workflow, evidence, verified observations, limitations, and future work for endpoint security monitoring with Wazuh.

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Document conventions

Verified observations, evidence placeholders, limitations, and future work are intentionally separated throughout this report.

1. Executive Summary

This project established a controlled virtual-lab workflow for endpoint security monitoring and threat-intelligence integration with Wazuh. The recorded work used an Intel-based MacBook Pro host, VMware Fusion, an Ubuntu Server 24.04.5 LTS Wazuh manager, and a Windows virtual machine with a Wazuh agent.

Windows telemetry included Sysmon and Windows Security events. Wazuh File Integrity Monitoring (FIM) was used to observe a file-addition event with a SHA-256 field. A manually maintained local SHA-256 indicator list supported controlled testing of custom Wazuh rule ID 100500.

The test used a harmless file and hash in a controlled virtual lab. No malware sample was downloaded or executed. Automated MalwareBazaar feed ingestion was not implemented. Results in this report distinguish recorded observations from untested capabilities and evidence that still needs to be inserted.

Key takeaway
The project demonstrated centralized telemetry collection, FIM visibility, and controlled custom-rule matching without claiming comprehensive malware detection.

2. Objective

The objective was to establish centralized endpoint monitoring that could collect security-relevant Windows events, identify file changes, and generate a controlled alert when a locally maintained SHA-256 indicator matched the test condition.

The project objectives were:

- **Deploy and access Wazuh.**
- **Connect the Windows agent.**
- **Collect Windows Security and Sysmon events.**
- **Configure and validate FIM.**
- **Test a custom hash-matching rule.**
- **Document the results and supporting evidence.**

3. Scope

In scope

Wazuh, Windows endpoint telemetry, Sysmon, Windows Security events, FIM, the local SHA-256 list, custom rule 100500, and controlled testing are in scope.

Out of scope

Production deployment, automated threat-intelligence feeds, malware execution, and comprehensive malware detection are out of scope.

Authorization

The work was performed in a controlled virtual lab.

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4. Environment and Tools

Component	Version / Details	Purpose
Intel-based MacBook Pro	Version/details not specified	Host computer.
VMware Fusion	Version not specified	Virtualization.
Ubuntu Server 24.04.5 LTS and Wazuh 4.14.9	Ubuntu Server 24.04.5 LTS; Wazuh 4.14.9	Centralized monitoring.
Windows virtual machine and Wazuh agent	Version not specified	Endpoint telemetry.
Sysmon	Version unspecified	Detailed Windows event collection.
Windows Security logs	Version not applicable	Security and authentication events.
Wazuh FIM	Version not specified	File-change monitoring.
/var/ossec/etc/lists/malware-sha256	Local SHA-256 indicator list	Manually maintained local indicators.
Custom Wazuh rule 100500	Rule ID 100500	Controlled matching alert.
GitHub, Markdown, Word, and PDF	Documentation formats and repository	Documentation and evidence.

5. Implementation

The following sequence records the implementation workflow. Each step separates purpose, recorded action, verification, and evidence that can be added later. No commands, exact settings, screenshots, alerts, or outputs are fabricated here.

Step 1. Deploy the Wazuh manager and access the dashboard

Purpose: Establish the central Wazuh monitoring environment.

Configuration or action: The recorded work used Ubuntu Server 24.04.5 LTS with Wazuh 4.14.9 under VMware Fusion on an Intel-based MacBook Pro.

Verification: Wazuh dashboard access was recorded as available.

Screenshot:

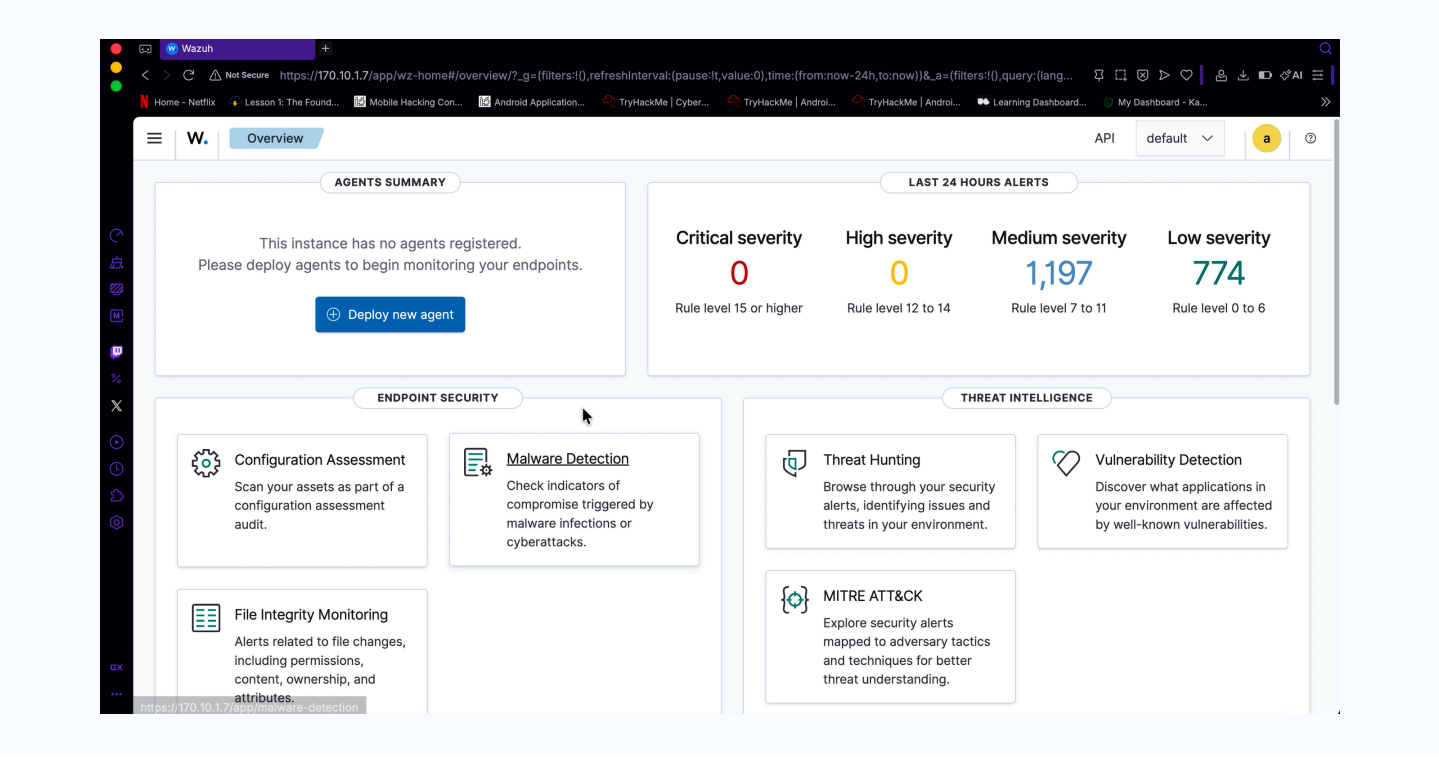


Figure 1. Wazuh manager dashboard access.

Step 2. Connect the Windows endpoint and verify agent status

Purpose: Connect the Windows virtual machine so endpoint telemetry can be received centrally.

Configuration or action: A Windows virtual machine with a Wazuh agent was connected to the manager.

Verification: The agent was observed active and reporting status in the recorded work.

Screenshot:

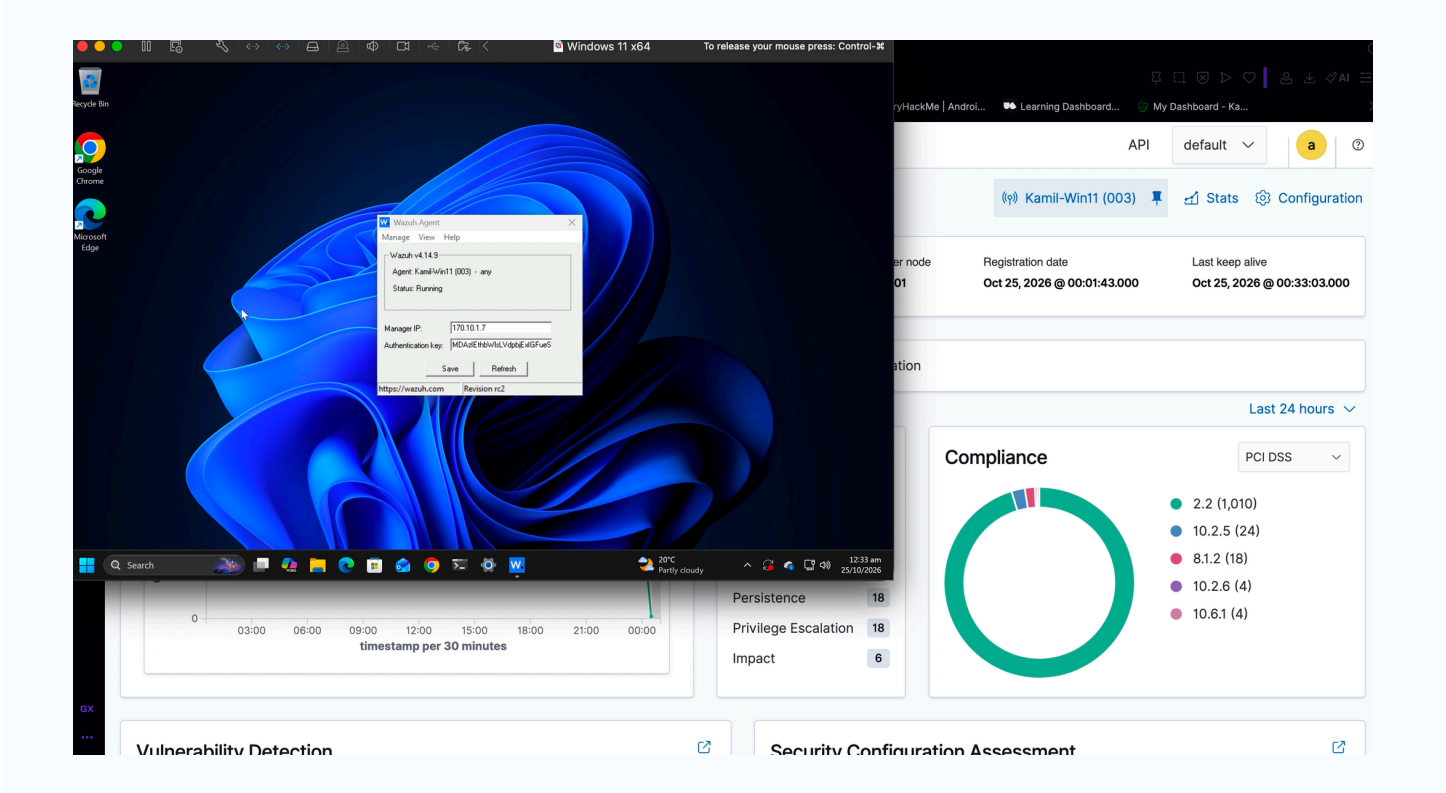


Figure 2. Windows agent active status.

Step 3. Install Sysmon and verify event collection

Purpose: Add detailed Windows event collection to the endpoint-monitoring workflow.

Configuration or action: Sysmon was installed in the Windows virtual machine as part of the recorded work.

Verification: Sysmon events were observed as available in Wazuh.

Screenshot:

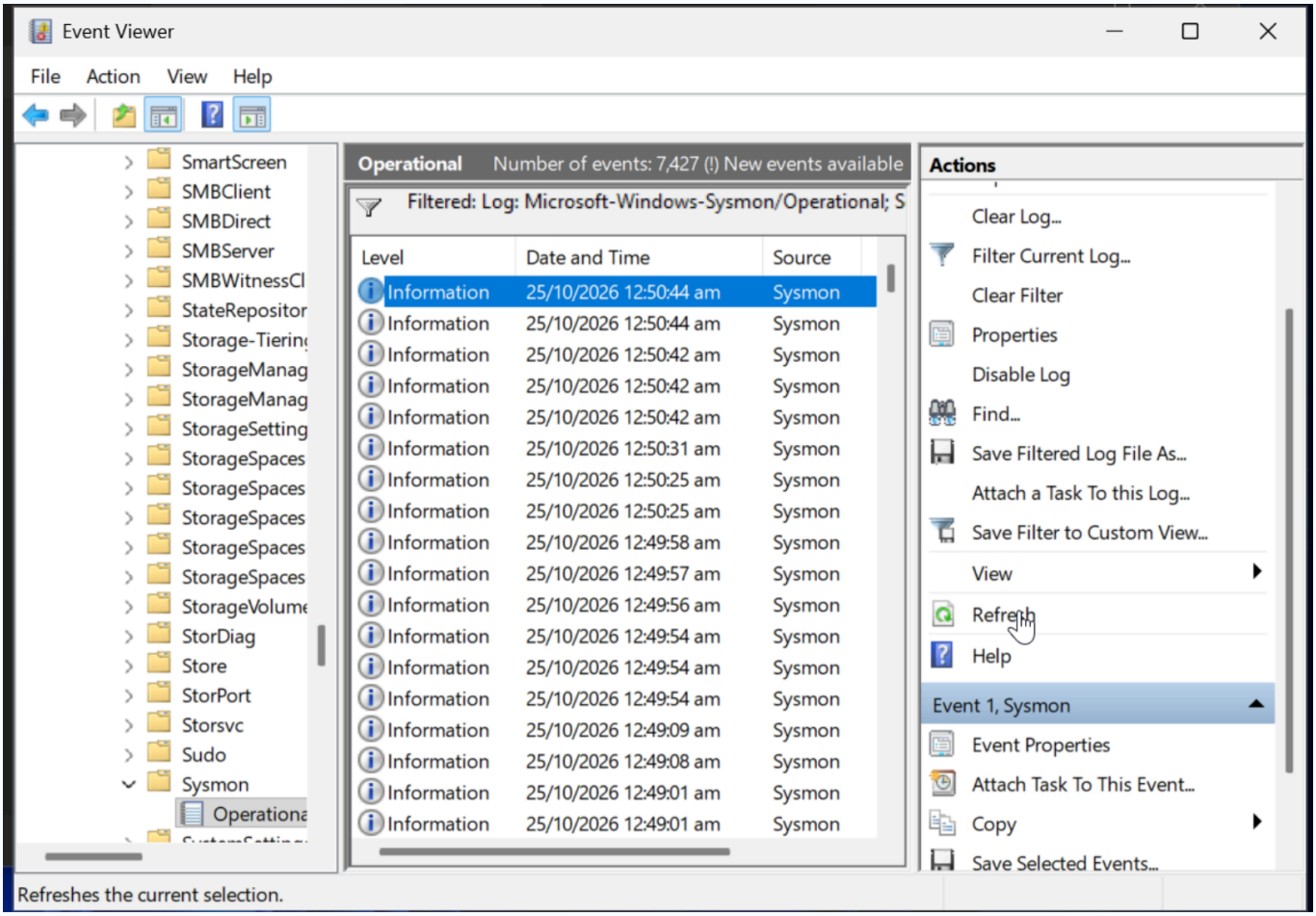


Figure 3. Sysmon event evidence.

Step 4. Verify Windows Security events in Wazuh

Purpose: Confirm that Windows Security telemetry is visible for security and authentication analysis.

Configuration or action: Windows Security events were reviewed through the Wazuh workflow.

Verification: Windows Security events were observed, including Event ID 4624 in the recorded work.

Screenshot:

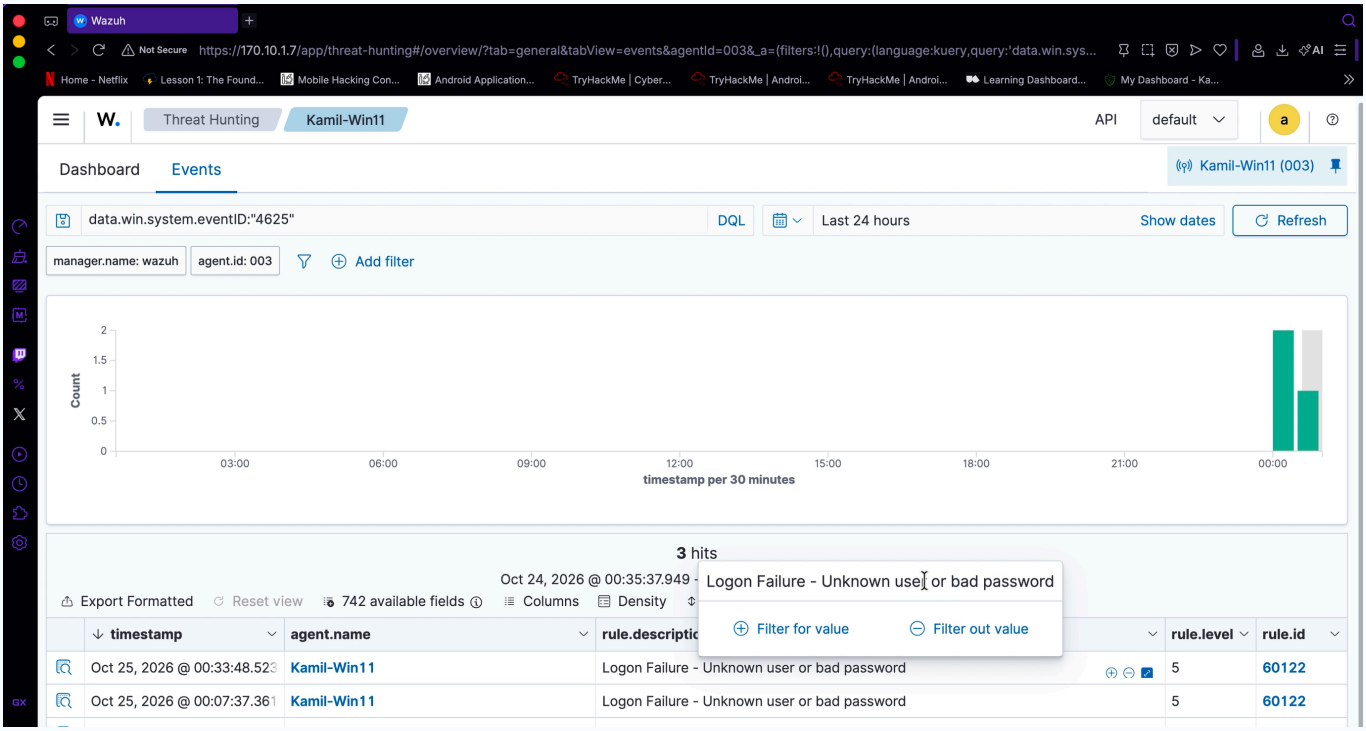


Figure 4. Windows Security event evidence.

Step 5. Configure FIM and observe a file-addition event with a SHA-256 field

Purpose: Monitor file changes and confirm that a file-addition event exposes a SHA-256 field.

Configuration or action: Wazuh File Integrity Monitoring was configured and used during the controlled lab workflow.

Verification: A file-addition event and a SHA-256 field were observed in the recorded work.

Screenshot:

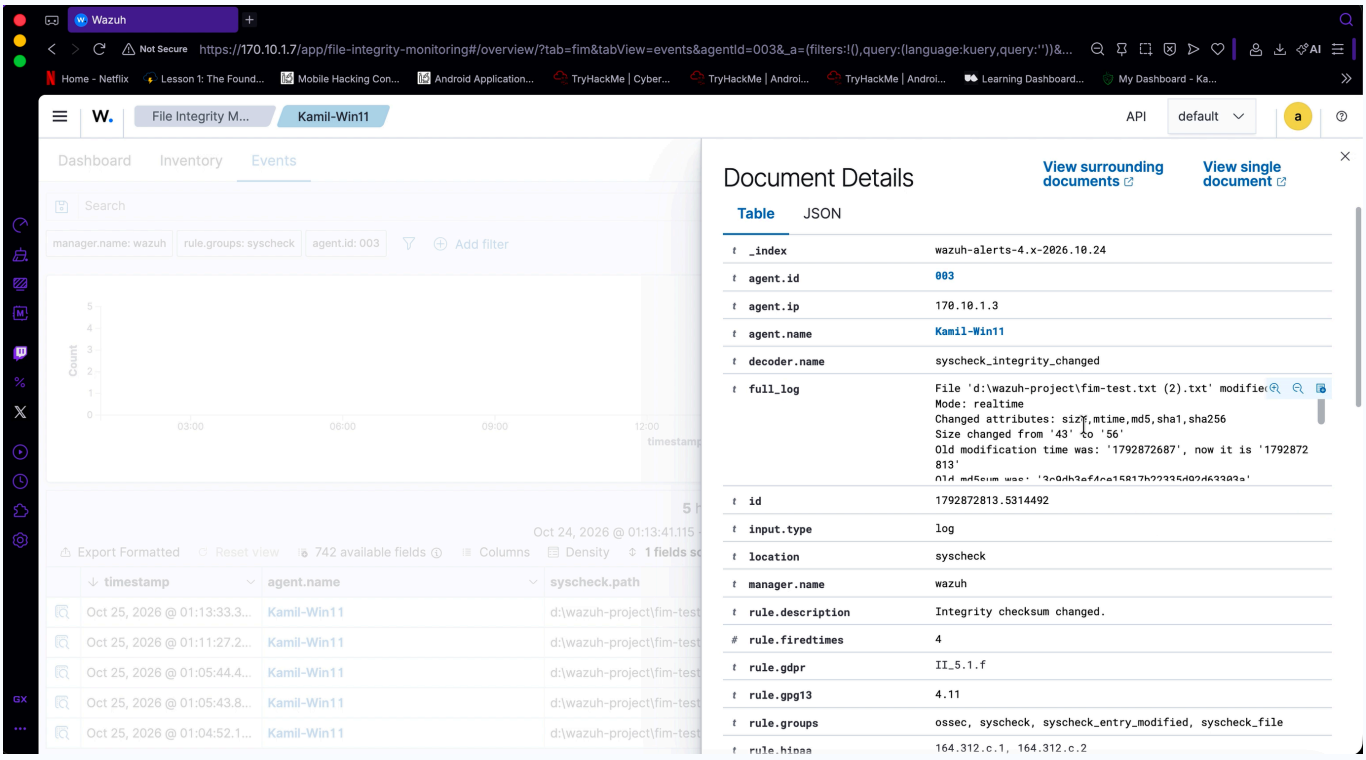


Figure 5. FIM file-addition event with a SHA-256 field.

6. Validation and Results

The table below records the observed outcomes from the controlled virtual-lab workflow. “Verified” is reserved for recorded observations; capabilities that were not performed remain untested.

Test	Expected Result	Observed Result	Status
Wazuh dashboard access	Dashboard accessible	Verified	Verified
Windows agent connectivity	Agent connects and reports status	Agent observed active	Verified
Sysmon collection	Sysmon events are available in Wazuh	Sysmon events observed	Verified
Windows Security collection	Windows Security events are available	Security events observed, including Event ID 4624 in the recorded work	Verified
FIM	File-addition event includes a SHA-256 field	File-addition event and SHA-256 field observed	Verified

Interpretation note
These are recorded observations. Untested capabilities must not be presented as verified, and a controlled test-hash match must not be interpreted as comprehensive malware detection.

7. Evidence Register

The evidence register identifies the intended files and the editable areas where screenshots can be inserted. The repository structure is `Endpoint-Security-Monitoring/evidence/`.

Evidence ID	Filename	Description
E-01	01-wazuh-dashboard-agent-active.png	Wazuh dashboard showing active agent
E-02	02-sysmon-event.png	Sysmon event evidence
E-03	03-windows-security-event.png	Windows Security event evidence
E-04	04-fim-sha256-event.png	FIM file-addition event with SHA-256 field

Evidence link policy: Create evidence links only after the corresponding files have actually been uploaded. Use this editable code-style pattern without presenting the files as already uploaded:

`https://github.com/mkbytes09/Project-with-Internee-pk/blob/main/Endpoint-Security-Monitoring/evidence/`

12. Conclusion

The project completed a controlled lab workflow for endpoint security monitoring with Wazuh. The recorded work demonstrated centralized Windows telemetry collection, Sysmon visibility, Windows Security event observation, FIM detection of a file addition with a SHA-256 field, and a controlled alert associated with custom rule 100500.

The result is a documented and editable foundation for evidence-backed monitoring. The manually maintained indicator list supported the test, while automated MalwareBazaar feed ingestion was not implemented. Production validation, comprehensive malware testing, and broader operational assessment remain future work.

Final statement
The demonstrated capabilities are meaningful within the stated controlled virtual-lab scope and should not be extended beyond the verified observations documented here.

13. References

Wazuh documentation: https://documentation.wazuh.com/

Wazuh ruleset documentation: https://documentation.wazuh.com/current/user-manual/ruleset/index.html

Microsoft Sysmon documentation: https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon

MalwareBazaar: https://bazaar.abuse.ch/

Repository reference:

GitHub - mkbytes09/Project-with-Internee-pk: Portfolio of projects completed during the Internee.pk program.

8. Findings and Limitations

Wazuh centralized endpoint telemetry from the Windows virtual machine. Sysmon and Windows Security events were observed in the recorded work, providing complementary visibility into detailed Windows activity and security or authentication events.

FIM detected a file addition and exposed a SHA-256 field. A controlled alert associated with custom rule 100500 was observed during testing. The indicator list was manually maintained, and automated MalwareBazaar ingestion was not implemented.

Limitations

- **Interpretation:** A test-hash match does not prove comprehensive malware detection.
- **Coverage:** Production performance and false-positive rates were not comprehensively assessed.
- **Threat intelligence:** The local indicator list was manually maintained rather than populated through automated feed ingestion.
- **Evidence:** Screenshot placeholders must be replaced with reviewed evidence before final publication.

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9. Security and Privacy

- **No malware:** No malware sample was downloaded or executed.
- **Safe reproduction:** Use harmless test files for reproduction.
- **Secrets:** Never publish passwords, tokens, private keys, or confidential logs.
- **Redaction:** Redact sensitive information in screenshots.
- **Scope:** Keep testing within the lab or an explicitly authorized scope.

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10. Lessons Learned

The project reinforced the value of centralized logging for bringing endpoint observations into one monitoring workflow. Sysmon increased visibility into Windows activity, while FIM provided a focused way to identify file changes and inspect event fields such as SHA-256.

The work also showed that event-field validation matters: a detection workflow should be checked against the fields actually exposed by the telemetry. Custom-rule testing should remain controlled and repeatable, and threat-intelligence matches must be interpreted carefully rather than treated as proof of broad malware-detection capability.